

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"Highlight the urban built-up area."
Task Category:	GROUNDING
Specialist Tool:	RS Text-Guided Region Grounding Specialist
Reliability Score:	58.2% (LOW TRUST)
Execution Time:	63.3 ms

1. Agent Synthesis & Visual-Language Answer

Text-Guided Region Grounding Output:

- 1. Target Entity Delineated: 'Built-Up / Urban Zone'.
- 2. Bounding Box Coordinates [ymin, xmin, ymax, xmax]:
 - Normalized: [0.3105, 0.3105, 0.3496, 0.3496]
 - Pixel Grid: [159, 159, 179, 179]
- 3. Grounded Region Extent: 0.15% of scene tile area.

2. Auditable Agent Pipeline Execution Trace

- **[Perception Layer]** Format & CRS validated. Mode: Single Optical Image (1 image(s)). Status: PASSED
- **[Query Engine]** Classified intent: 'GROUNDING'. Extracted entities: ['builtup']. Task: 'grounding'
- **[Mission Planner]** Selected tool: 'RS Text-Guided Region Grounding Specialist' with params {'target_entities': ['builtup']}. Generated 7 node task-graph.
- **[Specialist Tool Execution]** Executed specialist tool 'RS Text-Guided Region Grounding Specialist' with model confidence 40.0%.
- **[Evidence Fusion]** Aligned 2 evidence streams. Spatial Bounding Box aligned with query subject.
- **[Trust & Uncertainty]** Calculated Reliability Score: 58.2% (LOW TRUST). Cross-model agreement: 50.0%.
- **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 1 'Beyond Your Query' discovery item(s).
- **[Answer Generator]** Final evidence-linked response synthesized in 63.3 ms.